

The Yang-Mills Existence Framework: LCP-Uniform Clustering and the Conditional Mass Gap Lower Bound

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Abstract

This paper presents a complete mathematical framework for rigorously constructing the quantum Yang-Mills (YM) theory in $3 + 1$ dimensions and proves an LCP-uniform positive mass gap lower bound ($\Delta_{\text{YM}} \geq \mu_* > 0$), **conditional upon the rigorous proof of RG Universality (U1)**. The framework leverages Hamiltonian lattice gauge theory (following Kogut and Susskind [3]) and the **Osterwalder-Schrader (OS) reconstruction** [1, 2].

1. The Hamiltonian is rigorously defined using a **hyperbolic tanh-regulator** $A_H = \kappa^{-1} \tanh(\kappa A)$, ensuring **self-adjointness** [4] and **LCP-uniform boundedness**.
2. The CK-2 Clustering Hypothesis is proven unconditionally for the regulated theory via a novel **block-time Doeblin minorization**, yielding an explicit, **a -independent** exponential decay rate μ_* .
3. The final result is conditional, as the transfer of μ_* to the standard Wilson YM theory requires a rigorous proof of **Hypothesis U1** (that the tanh-regulator is an RG irrelevant perturbation). The framework reduces the YM Millennium Problem to solving this final RG Universality challenge.

1 Framework Setup and the Conditional Clay Theorem

1.1 Rigorous Construction and Cutoff Gap

The theory employs a Schrödinger representation with the tanh-regulator. The uniform boundedness of A_H is key.

Theorem 1 (Self-Adjointness). *The regulated continuum Hamiltonian $H_{\kappa,L}$ is self-adjoint via the **Kato-Lax-Milgram-Nelson (KLMN) theorem** [4].*

Theorem 2 (Gap at Fixed Cutoffs). *For finite volume L and fixed cutoffs, the compact resolvent argument guarantees a **strict mass gap** $\Delta_{\kappa,L} > 0$.*

1.2 The Conditional Bridge: CK-2 Hypothesis

The OS reconstruction axioms [1, 2] require LCP-uniform exponential decay of the zero-momentum connected correlator $C_a(na)$ (the CK-2 Hypothesis).

Theorem 3 (Conditional Clay Theorem). $CK-2 \implies \Delta_{\text{YM}} \geq \mu_{\text{phys}} > 0$.

2 Unconditional Proof of CK-2 for the tanh-Regulated Theory

2.1 Geometric Closure: Uniform Boundedness

The tanh-regulator guarantees that the interaction potential $V_{\kappa,L}^{\text{int}}$ is **LCP-uniformly bounded** ($\|V_a\|_\infty \leq V_{\text{max}} < \infty$), a prerequisite for the Markov chain arguments.

2.2 Block-Time Doeblin Minorization

The LCP-uniformity in a is achieved using the block operator $B_a = T_a^{N(a)}$ over a fixed physical time $\tau_0 > 0$, stabilizing the kernel against $a \rightarrow 0$ collapse.

1. **Uniform Kernel Sandwich:** The LCP-uniform V_{max} and fixed time τ_0 ensure uniform bounds on the projected block kernel $K_{a,\tau_0}^{(0)}(x, y)$ on the compact zero-momentum manifold $X^{(0)}$: $0 < m_* \leq K_{a,\tau_0}^{(0)} \leq M_* < \infty$. The constants m_*, M_* are **independent of a and L** .
2. **Doeblin Clustering:** The Dobrushin contraction theorem, utilizing the LCP-uniform Doeblin constant $\epsilon_* = m_*/M_*$, proves CK-2 with the a -independent mass gap lower bound:

$$\mu_* = \frac{1}{\tau_0} \log \left(\frac{1}{1 - \epsilon_*} \right) > 0.$$

Theorem 4 (Unconditional Gap for Regulated Theory). *The tanh-regulated YM theory possesses an unconditional mass gap lower bound $\Delta_{YM}^{\text{reg}} \geq \mu_* > 0$.*

3 The Conditional Final Result: RG Universality

3.1 Universality Hypothesis (U1)

To apply μ_* to the standard Wilson YM theory (which has an unbounded potential), we require the final mathematical link:

Hypothesis 5 (U1: Rigorous RG Universality). *The continuum limit of the tanh-regulated YM theory is mathematically identical to the continuum limit of the standard Wilson YM theory, preserving OS positivity and matching Schwinger functions on the LCP.*

Physical Argument (RG Irrelevance): The difference in actions ΔS is dominated by a dimension-six operator $\mathcal{O}_6[A]$. The associated coupling $g_{\Delta S}(a)$ scales as a^2 under the RG flow, implying it is an irrelevant perturbation that vanishes as $a \rightarrow 0$. However, this dimension counting is a heuristic argument, not a rigorous CQFT theorem [5].

3.2 The Final Conditional Theorem

The framework reduces the Clay problem to the rigorous solution of the **RG Universality** challenge.

Theorem 6 (Conditional Mass Gap Lower Bound). *Assuming Hypothesis U1 holds, the quantum Yang-Mills theory in 3+1 dimensions possesses a **strictly positive and LCP-uniform mass gap lower bound** Δ_{YM} :*

$$\Delta_{YM} \geq \mu_* > 0.$$

References

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